AFKEBOOKS
SINCE 2013

or hantavirus cardiopulmonary syndrome) share epidemiologic and clinical features with leptospirosis. Dual infections have been reported. In this light, it is advisable to conduct serologic testing for rickettsiae, dengue virus, and hantavirus when leptospirosis is suspected. When bleeding is detected, dengue hemorrhagic fever and other viral hemorrhagic fevers, including hantavirus infection, yellow fever, Rift Valley fever, filovirus infections, and Lassa fever, should be considered.

TREATMENT

Leptospirosis

Severe leptospirosis should be treated with IV penicillin (Table 184-1) as soon as the diagnosis is considered. *Leptospira* are highly susceptible to a broad range of antibiotics, including the β-lactam antibiotics, cephalosporins, aminoglycosides, and macrolides, but are not susceptible to vancomycin, rifampicin, metronidazole, and chloramphenicol. Early intervention may prevent the development of major organ-system failure or lessen its severity. Although studies supporting antibiotic therapy have produced conflicting results, clinical trials are difficult to perform in settings where patients frequently present for medical care with late stages of disease. Antibiotics are less likely to benefit patients in whom organ damage has already occurred. Two open-label randomized studies comparing penicillin with parenteral cefotaxime, parenteral ceftriaxone, and doxycycline showed no significant differences among the antibiotics with regard to complications and mortality risk. Thus ceftriaxone, cefotaxime, or doxycycline is a satisfactory alternative to penicillin for the treatment of severe leptospirosis. Antimicrobial susceptibility testing is not routine practice in individual cases of leptospirosis; to date, however, antibiotic resistance has not been reported in isolates from patients or the environment.

In mild cases, oral treatment with doxycycline, azithromycin, ampicillin, or amoxicillin is recommended. In regions where rickettsial diseases are coendemic, doxycycline or azithromycin is the drug of choice. In rare instances, a Jarisch-Herxheimer reaction develops within hours after the initiation of antimicrobial therapy.

Aggressive supportive care for leptospirosis is essential and can be life-saving. Patients with nonoliguric renal dysfunction require aggressive fluid and electrolyte resuscitation to prevent dehydration and precipitation of oliguric renal failure. Peritoneal dialysis or hemodialysis should be provided to patients with oliguric renal failure. Rapid initiation of hemodialysis has been shown to reduce mortality risk and typically is necessary only for short periods. Patients with pulmonary hemorrhage may have reduced pulmonary compliance (as seen in ARDS) and may benefit from mechanical ventilation with low tidal volumes to avoid high ventilation pressures. Evidence is contradictory for the use of glucocorticoids and desmopressin as adjunct therapy for pulmonary involvement associated with severe leptospirosis.

TABLE 184-1 Treatment and Chemoprophylaxis of Leptospirosis in Adults[a]

INDICATION	REGIMEN
Treatment	
Mild leptospirosis	Doxycycline[b] (100 mg PO bid) *or* Amoxicillin (500 mg PO tid) *or* Ampicillin (500 mg PO tid)
Moderate/severe leptospirosis	Penicillin (1.5 million units IV or IM q6h) *or* Ceftriaxone (2 g/d IV) *or* Cefotaxime (1 g IV q6h) *or* Doxycycline[b] (loading dose of 200 mg IV, then 100 mg IV q12h)
Chemoprophylaxis	
	Doxycycline[b] (200 mg PO once a week) *or* Azithromycin (250 mg PO once or twice a week)

[a]All regimens are given for 7 days. [b]Doxycycline should not be given to pregnant women or children. [c]The efficacy of doxycycline prophylaxis in endemic or epidemic settings remains unclear. Experiments in animal models and a cost-effectiveness model indicate that azithromycin has a number of characteristics that may make it efficacious in treatment and prophylaxis.

PROGNOSIS

Most patients with leptospirosis recover. However, post-leptospirosis symptoms, mainly of a depression-like nature, may occur and persist for years after the acute disease. Mortality rates are highest among patients who are elderly and those who have severe disease (pulmonary hemorrhage, Weil's syndrome). Leptospirosis during pregnancy is associated with high fetal mortality rates. Long-term follow-up of patients with renal failure and hepatic dysfunction has documented good recovery of renal and hepatic function.

PREVENTION

Individuals who may be exposed to *Leptospira* through their occupations or their involvement in recreational freshwater activities should be informed about the risks. Measures for controlling leptospirosis include avoidance of exposure to urine and tissues from infected animals through proper eyewear, footwear, and other protective equipment. Targeted rodent control strategies could be considered.

Vaccines for agricultural and companion animals are generally available, and their use should be encouraged. The veterinary vaccine used in a given area should contain the serovars known to be present in that area. Unfortunately, some vaccinated animals still excrete leptospires in their urine. Vaccination of humans against a specific serovar prevalent in an area has been undertaken in some European and Asian countries and has proved effective. Although a large-scale trial of vaccine in humans has been reported from Cuba, no conclusions can be drawn about efficacy and adverse reactions because of insufficient details on study design. The efficacy of chemoprophylaxis with doxycycline (200 mg once a week) or azithromycin (in pregnant women and children) is being disputed, but focused pre- and postexposure administration is indicated in instances of well-defined short-term exposure (Table 184-1).

FURTHER READING

Adler A: *Leptospira and Leptospirosis*, 1st ed. Berlin Heidelberg, Springer-Verlag, 2015.

de Vries SG et al: Leptospirosis among returned travelers: A GeoSentinel Site Survey and Multicenter Analysis–1997–2016. Am J Trop Med Hyg 99:127, 2018.

Haake DA, Levett PN: Leptospirosis in humans. Curr Top Microbiol Immunol 387:65, 2015.

van Samkar A et al: Suspected leptospiral meningitis in adults: Report of four cases and review of the literature. Neth J Med 73:464, 2015.

Vincent AT et al: Revisiting the taxonomy and evolution of pathogenicity of the genus Leptospira through the prism of genomics. PLoS Negl Trop Dis 13:e0007270, 2019.

185 Relapsing Fever and *Borrelia miyamotoi* Disease

Alan G. Barbour

Relapsing fever is caused by infection with any of several species of *Borrelia* spirochetes. Physicians in ancient Greece distinguished relapsing fever from other febrile disorders by its characteristic clinical presentation: two or more fever episodes separated by varying periods of well-being. In the nineteenth century, relapsing fever was one of the first diseases to be associated with a specific microbe by virtue of its characteristic laboratory finding: the presence of large numbers of spirochetes of the genus *Borrelia* in the blood.

AFKEBOOKS
SINCE 2013

The host responds with systemic inflammation that results in an illness ranging from a flulike syndrome to sepsis. Other manifestations are the consequences of central nervous system (CNS) involvement and disordered hemostasis. Antigenic variation of the spirochetes' surface proteins accounts for the infection's relapsing course. Acquired immunity follows the serial development of antibodies to each of the several variants appearing during an infection. Treatment with antibiotics results in rapid cure but at the risk of a moderate to severe Jarisch-Herxheimer reaction.

Louse-borne relapsing fever (LBRF) caused large epidemics well into the twentieth century and currently occurs in northeastern Africa and among migrants from that area. At present, however, most cases of relapsing fever are tick-borne in origin. Sporadic cases and small outbreaks are focally distributed on most continents, with Africa and Central Asia most affected. In North America, the majority of reports of relapsing fever have been from the western United States and Canada.

Another member of the genus, *Borrelia miyamotoi*, causes an acute febrile illness with nonspecific constitutional symptoms and occasionally meningoencephalitis in the same geographic distribution as Lyme disease **(Chap. 186)** in Eurasia and North America.

ETIOLOGIC AGENT

Coiled, thin microscopic filaments that swim in one direction and then coil up before heading in another were first observed in the blood of patients with relapsing fever in the 1880s. These microbes were categorized as spirochetes and assigned to the genus *Borrelia*. The breakthrough cultivation medium was rich in ingredients, ranging from simple (e.g., *N*-acetylglucosamine) to more complex (e.g., serum). The limited biosynthetic capacity of *Borrelia* cells is accounted for by a genome content one-quarter that of *Escherichia coli*.

Like other spirochetes, the helix-shaped *Borrelia* cells have two membranes, the outer of which is more loosely secured than in other double-membrane bacteria, such as *E. coli*. As a consequence, fixed organisms with damaged membranes can assume a variety of morphologies in smears and histologic preparations. The flagella of spirochetes run between the two membranes and are not on the cell surface. Although technically gram-negative, the 10- to 20-μm-long *Borrelia* cells, with a diameter of 0.2–0.3 μm, are too narrow to be seen by microscopy of Gram-stained slides.

EPIDEMIOLOGY

The several species of *Borrelia* that cause relapsing fever have restricted geographic distributions **(Table 185-1)**. The exception is *Borrelia recurrentis*, which is also the only species transmitted by an insect. LBRF is acquired from a body louse (*Pediculus humanus corporis*), or possibly a head louse (*Pediculus capitis*), with humans serving as the reservoir. Acquisition occurs not from the bite itself but from either rubbing the insect's feces into the bite site with the fingers in response to irritation or inoculation into the conjunctivae or a wound. Although LBRF transmission is currently limited to Ethiopia, Eritrea, and Somalia, the disease has had a global distribution in the past, and that potential remains. Epidemics of LBRF, often in association with typhus, can occur under circumstances of famine, refugee migration, war, and pervasive homelessness. Transmission of LBRF can occur in camps of migrants at a distance from their home countries.

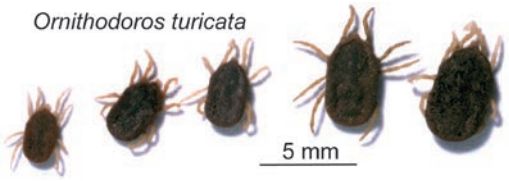


FIGURE 185-1 ***Ornithodoros turicata*** **soft ticks** of different ages.

All other known species of relapsing fever agents are tick-borne—in most cases, by soft ticks of the genus *Ornithodoros* **(Fig. 185-1)**. Tick-borne relapsing fever (TBRF) is found on most continents but is absent in tropical or arctic environments. For most species, the reservoirs of infection are small to medium-sized mammals, usually rodents but sometimes pigs and other domestic animals living around human habitats. However, one species, *Borrelia duttonii* in sub-Saharan Africa, is largely maintained by tick transmission between human hosts. In North America, TBRF occurs as single cases or small case clusters through transient exposure of persons to infested buildings or caves where mammals have nests or sleep in less populated areas. The two main *Borrelia* species involved in North America are *Borrelia hermsii* in the mountainous west and *Borrelia turicatae* in arid southwestern and south-central regions. The soft tick vectors typically feed for no more than 30 min, usually while the victim is sleeping. Transovarial transmission from one generation of ticks to the next means that infection risk may persist in an area long after incriminated mammalian reservoirs have been removed.

Borrelia miyamotoi belongs to the same clade as relapsing fever species but instead is transmitted to humans from other mammals by hard ticks (e.g., *Ixodes scapularis* in the eastern United States) that also transmit Lyme disease, babesiosis, anaplasmosis, and a viral encephalitis. *B. miyamotoi* is acquired through outdoor activities and through contact with ticks in forested and shrubby areas during recreation, work, or activities around the home, similarly to Lyme disease **(Chap. 186)**. Among residents of most areas where *B. miyamotoi* and *Borreliella* (also called *Borrelia*) *burgdorferi* coexist, the prevalence of antibodies to the former is about one-third of that to the latter. In contrast to *B. burgdorferi*, the transmission of *B. miyamotoi* to the host begins soon after the tick begins to feed.

PATHOGENESIS AND IMMUNITY

TBRF spirochetes enter the body in the tick's saliva with the onset of feeding. From an inoculum of a few cells, the spirochetes proliferate

TABLE 185-1 Relapsing Fever *Borrelia* Species, by Geographic Region, Vector, and Primary Reservoir

SPECIES	REGION(S)	ARTHROPOD VECTOR(S)	PRIMARY RESERVOIR
B. crocidurae	Africa	*Ornithodoros erraticus, Ornithodoros sonrai* (soft ticks)	Mammals
B. duttonii	Africa	*O. moubata*	Humans
B. hermsii	North America	*O. hermsi*	Mammals
B. hispanica	Europe, North Africa	*O. erraticus*	Mammals
B. johnsonii	North America	*Carios kellyi* (soft ticks)	Bats
B. kalaharica	Africa	*O. savignyi*	Mammals
B. mazzottii	Mexico, Central America	*O. talaje*	Mammals
B. miyamotoi	Eurasia, North America	*Ixodes* species (hard ticks)	Mammals
B. persica	Eurasia	*O. tholozani*	Mammals
B. recurrentis	Africa, global[a]	*Pediculus humanus corporis* (human body louse)	Humans
B. turicatae	North America	*O. turicata*	Mammals
B. venezuelensis	Central and South America	*O. rudis*	Mammals

[a]Although transmission is currently limited to Ethiopia and adjacent countries, *B. recurrentis* infection has had a global distribution in the past, and that potential remains.

AFKEBOOKS
SINCE 2013

in the blood, doubling every 6 h to numbers of 10^6–10^7/mL or more. *Borrelia* species are extracellular pathogens; their presence inside cells connotes dead bacteria after phagocytosis. Binding of the spirochetes to erythrocytes leads to aggregation of red blood cells, their sequestration in the spleen and liver, and hepatosplenomegaly and anemia. A bleeding disorder is probably the consequence of thrombocytopenia, impaired hepatic production of clotting factors, and/or blockage of small vessels by aggregates of spirochetes, erythrocytes, and platelets. Some species are neurotropic and enter the brain, where they are comparatively sheltered from host immunity. Relapsing fever spirochetes can cross the maternal-fetal barrier and cause placental damage and inflammation, leading to intrauterine growth retardation and congenital infection.

Although *Borrelia* species do not have potent exotoxins or a lipopolysaccharide endotoxin, they have abundant lipoproteins that activate Toll-like receptors on host cells, which leads to a proinflammatory process similar to that in endotoxemia, with elevations of tumor necrosis factor α, interleukin 6, and interleukin 8 concentrations.

IgM antibodies specific for the serotype-defining surface lipoprotein appear after a few days of infection and soon reach a concentration that causes lysis of bacteria in the blood through either direct bactericidal action or opsonization. The release of lipoproteins and other bacterial products from dying bacteria provokes a "crisis," during which there can be an increase in temperature, hypotension, and other signs of shock. A similar phenomenon occurring in some patients soon after the initiation of antibiotic treatment is characterized by an abrupt worsening of the patient's condition, which is called a Jarisch-Herxheimer reaction (JHR).

CLINICAL MANIFESTATIONS

Relapsing fever presents with the sudden onset of fever. Febrile periods are punctuated by intervening afebrile periods of a few days; this pattern occurs at least twice. The patient's temperature is ≥39°C and may be as high as 43°C. The first fever episode often ends in a crisis lasting ~15–30 min and consisting of rigors, a further elevation in temperature, and increases in pulse and blood pressure. The crisis phase is followed by profuse diaphoresis, falling temperature, and hypotension, which usually persist for several hours. In LBRF, the first episode of fever is unremitting for 3–6 days; it is usually followed by a single milder episode. In TBRF, multiple febrile periods last 1–3 days each. In both forms, the interval between fevers ranges from 4 to 14 days, sometimes with symptoms of malaise and fatigue.

The symptoms that accompany the fevers are usually nonspecific. Headache, neck stiffness, arthralgia, myalgia, and vomiting may accompany the first and subsequent febrile episodes. An enlarging spleen and liver cause abdominal pain. A nonproductive cough is common during LBRF and—in combination with fever and myalgias—may suggest influenza. Acute respiratory distress syndrome may occur during TBRF.

On physical examination, the patient may be delirious or apathetic. There may be body lice in the patient's clothes or signs of insect bites. In regions with *B. miyamotoi* infection, a hard tick may be embedded in the skin. Epistaxis, petechiae, and ecchymoses are common during LBRF but not in TBRF. Splenomegaly or spleen tenderness is common in both forms of relapsing fever. The majority of patients with LBRF and ~10% of patients with TBRF have discernible hepatomegaly.

Localizing neurologic findings are more common in TBRF than in LBRF. In North America, *B. turicatae* infection has neurologic manifestations more often than *B. hermsii* infection. Meningoencephalitis can result in residual hemiplegia or aphasia. Myelitis and radiculopathy may develop. Unilateral or bilateral Bell's palsy and deafness from seventh or eighth cranial nerve involvement are the most common forms of cranial neuritis and typically present in the second or third febrile episode, not the first. Visual impairment from unilateral or bilateral iridocyclitis or panophthalmitis may be permanent. In LBRF, neurologic manifestations such as altered mental state or stiff neck are thought to be secondary to systemic inflammation rather than to direct invasion of the nervous system.

Myocarditis appears to be common in both forms of relapsing fever and accounts for some deaths. Most commonly, myocarditis is evidenced by gallops on cardiac auscultation, a prolonged QT_c interval, and cardiomegaly and pulmonary edema on chest radiography.

General laboratory studies are not specific. Mild to moderate normocytic anemia is common, but frank hemolysis and hemoglobinuria do not develop. Leukocyte counts are usually in the normal range or only slightly elevated, and leukopenia can occur during the crisis. Platelet counts can fall below 50,000/μL. C-reactive protein and procalcitonin levels are elevated. Laboratory evidence of hepatitis can be found, with elevated serum concentrations of unconjugated bilirubin and aminotransferases; the prothrombin and partial thromboplastin times may be moderately prolonged.

Analysis of the cerebrospinal fluid (CSF) is indicated in cases of suspected relapsing fever with signs of meningitis or meningoencephalitis. The presence of mononuclear pleocytosis and mildly to moderately elevated protein levels justifies intravenous antibiotic therapy in TBRF.

The manifestations and course of *B. miyamotoi* disease are not as distinctive as those of relapsing fever. The most common presentation is fever without respiratory symptoms starting 1–2 weeks after a tick bite. Patients have been hospitalized with a presumptive diagnosis of undifferentiated sepsis. Meningoencephalitis with spirochetes in the CSF was documented in immunodeficient adults but may also occur in immunocompetent individuals. If the patient has coexisting early Lyme disease, there may be erythema migrans, the localized skin rash.

DIAGNOSIS

Relapsing fever should be considered in a patient with the characteristic fever pattern and a history of recent exposure—i.e., within 1–2 weeks before illness onset—to body lice or soft-bodied ticks in geographic areas with documented current or past transmission. Because of the longevity of the ticks and the transovarial transmission of the pathogen in the ticks, a case of relapsing fever may be diagnosed many years after the last case reported in that locale. The lice may be on the clothes of a migrant. While the risks for *B. miyamotoi* disease are similar to those for Lyme disease, prompt removal of an embedded tick after the exposure may not reduce the risk of infection of this pathogen.

With the exception of *B. miyamotoi* infection, the bedrock for laboratory diagnosis of LBRF and TBRF remains direct detection of the spirochetes by microscopy of the blood. Manual differential counts of white blood cells by Wright stain usually reveal spirochetes in thin blood smears if their concentration is ≥10^5/mL and several oil-immersion fields are examined **(Fig. 185-2)**. But the preferred stains are Giemsa-Wright or Giemsa alone. The density of *B. miyamotoi* in the blood is not high enough for use of a blood smear alone for diagnosis. For LBRF and TBRF, the preferred time to obtain a blood specimen is between the fever's onset and its peak. Lower concentrations of spirochetes may be revealed by a thick blood smear that is treated with 0.5% acetic acid before staining. An alternative is a wet mount of anticoagulated blood mixed with saline and examined by phase-contrast or dark-field microscopy for motile spirochetes.

Polymerase chain reaction (PCR) and similar DNA amplification procedures are increasingly used for examination of blood or CSF in cases of suspected relapsing fever. PCR may reveal circulating spirochetes between febrile episodes. PCR is the preferred procedure for direct detection of *B. miyamotoi* in blood or CSF.

Culture of blood or CSF in Barbour-Stoenner-Kelly broth medium or equivalent is an option for isolation of *Borrelia* species. However, few laboratories offer this service. An alternative for tick-borne *Borrelia* species, but not *B. recurrentis*, is inoculation of blood or CSF into severe combined immunodeficient mice and examination of the animal's blood after a few days.

Options for serologic confirmation of infection are limited, and results may be misleading. Whole cell-based assays, such as enzyme-linked immunosorbent assay (ELISA) and immunoblot, and the C6-peptide ELISA for Lyme disease may be positive in relapsing fever or *B. miyamotoi* disease through antigenic cross-reactivities among these spirochetes. A commercially available assay based on GlpQ, a protein antigen of all relapsing fever *Borrelia* species (including *B. miyamotoi*) but not of any Lyme disease species, has better specificity, but commonly is negative at a time when a blood smear or PCR assay

AFKEBOOKS
SINCE 2013

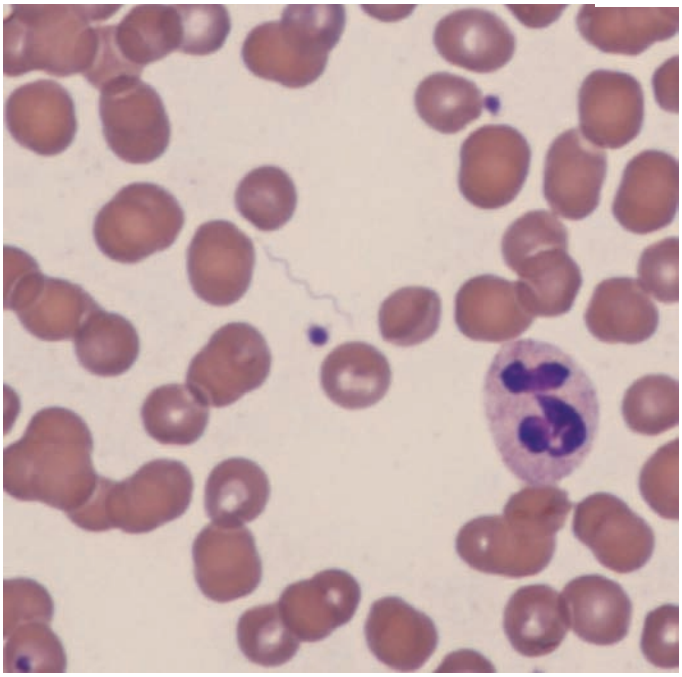


FIGURE 185-2 Photomicrograph of tick-borne relapsing fever spirochete (*Borrelia turicatae*) in a Giemsa-Wright–stained thin blood smear. Included in the figure are a polymorphonuclear leukocyte and two platelets.

would be positive. The results of current GlpQ-based assays cannot be used to differentiate between different *Borrelia* species as to etiology.

DIFFERENTIAL DIAGNOSIS

Depending on the patient's history of residential, occupational, travel, and recreational exposures, the differential diagnosis of relapsing fever includes one or more of the following infections that feature either periodicity in the fever pattern or an extended single febrile period with nonspecific constitutional symptoms: Colorado tick fever, Rocky Mountain spotted fever and other rickettsioses, ehrlichiosis, anaplasmosis, tick-borne viral infection, rat bite fever, and babesiosis in North America, Europe, Russia, and northeastern Asia. Elsewhere in the Americas and Asia and in most of Africa, malaria, typhoid fever, typhus and other rickettsioses, dengue, and leptospirosis may also be considered. There may be co-infections of malaria, typhus, or typhoid with TBRF or LBRF. *B. miyamotoi* infection may coexist with Lyme disease, anaplasmosis, or babesiosis in North America.

TREATMENT

Relapsing Fever

Penicillin and tetracyclines have been the antibiotics of choice for relapsing fever for several decades. Erythromycin and chloramphenicol have been long-standing alternative choices. There is no evidence of acquired resistance to these antibiotics. *Borrelia* species are also susceptible to second- and third-generation cephalosporins. These spirochetes are also relatively resistant to rifampin, sulfonamides, and aminoglycosides. Spirochetes are no longer detectable in the blood within a few hours after the first dose of an effective antibiotic.

Under conditions of limited resources or in the midst of an epidemic, a single dose of antibiotic usually suffices for successful treatment of LBRF **(Fig. 185-3)**. For adults, a single dose of oral tetracycline (500 mg), oral doxycycline (200 mg), or intramuscular penicillin G procaine (800,000 units) is effective. The corresponding doses for children are oral tetracycline at 12.5 mg/kg, oral doxycycline at 5 mg/kg, and intramuscular penicillin G procaine at 200,000–400,000 units. When an adult patient is stuporous or nauseated, the intravenous dose of tetracycline is 250–500 mg. Tetracyclines are contraindicated in pregnant and nursing women and in children <9 years old; for individuals in these groups who are allergic to penicillin, oral erythromycin (500 mg for adults and 12.5 mg/kg for children) is an alternative. While there is little reported experience with other macrolides, such as azithromycin, these are likely to be as effective as erythromycin.

But there are shortcomings of single-dose therapies for LBRF. With penicillin alone, recurrence may occur in up to 20% of patients, and the frequency of JHR was higher after tetracycline than penicillin. For treatment of LBRF in adults in Ethiopia, a regimen that reduces rates of both recurrence and JHR has been a single dose of 400,000 units of intramuscular penicillin G procaine

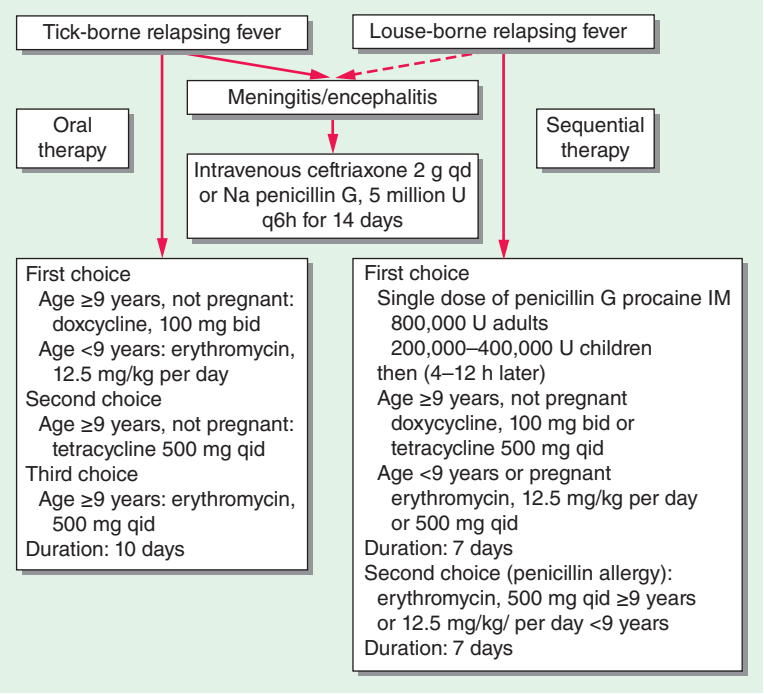


FIGURE 185-3 Algorithm for treatment of relapsing fever. If it is not known whether the patient has tick-borne or louse-borne relapsing fever, the patient should be treated for the tick-borne form. The *dashed line* indicates that central nervous system invasion in louse-borne relapsing fever is uncommon.

AFKEBOOKS
SINCE 2013

followed several hours later or the next day by doxycycline (100 mg orally twice daily) or tetracycline (500 mg or 12.5 mg/kg orally every 6 h) for 7 days.

The accumulated anecdotal reports on TBRF therapy indicate a recurrence rate of ≥20% after single-dose treatment, plausibly due to the propensity of some tick-borne species to invade the CNS. Accordingly, multiple antibiotic doses are recommended. The preferred treatment for adults is a 10-day course of doxycycline (100 mg twice daily) or tetracycline (500 mg or 12.5 mg/kg orally every 6 h). When tetracyclines are contraindicated, the alternative is erythromycin (500 mg or 12.5 mg/kg orally every 6 h) for 10 days. If a β-lactam antibiotic is given and CNS involvement is confirmed or suspected, it is preferably administered intravenously rather than orally. For adults, the regimen is penicillin G (5 million units IV every 6 h) or ceftriaxone (2 g IV daily) for 10–14 days. Under conditions of limited resources or when CNS involvement is not suspected, oral penicillin V potassium (500 mg or 12.5 mg/kg every 6–8 h) for 10 days is used.

The JHR during treatment of relapsing fever can be severe and may end in death if precautions are not in place for close monitoring for at least 24 h and with provision of parenteral cardiovascular and volume support as needed. Apprehension, rigors, fever, and hypotension occur within 1–3 h of initiation of antibiotic treatment and may be accompanied by a further decrease in the platelet count. The incidence of the JHR is 20–60% in LBRF after the first antibiotic dose. JHR may also be encountered when a patient with unsuspected relapsing fever is treated with other types of antibiotics, such ciprofloxacin, that have suboptimal effects.

Experience with the treatment of *B. miyamotoi* infection is limited, but this organism likely has the same antibiotic susceptibilities as other *Borrelia* species. Therapy for *B. miyamotoi* disease follows the guidelines for Lyme disease. This would include parenteral therapy for CNS involvement. In absence of contraindications, doxycycline (100 mg twice daily) is the preferred choice for uncomplicated *B. miyamotoi* infection, because of the antibiotic's efficacy for anaplasmosis and Lyme disease. If JHR occurs, it is generally milder than is observed in relapsing fever.

PROGNOSIS

The mortality rates for untreated LBRF and TBRF are in the ranges of 10–70% and 4–10%, respectively, and are largely determined by coexisting conditions, such as malnutrition, dehydration, or another infection. With prompt antibiotic treatment, the mortality rate is 2–5% for LBRF and <2% for TBRF. Features associated with a poor prognosis include concurrence with malaria, typhus, or typhoid; pregnancy; stupor or coma on admission; diffuse bleeding; poor liver function; myocarditis; and bronchopneumonia. The mortality rate from the JHR in LBRF, in the absence of adequate monitoring and resuscitation measures, is ~5%. Relapsing fever during pregnancy frequently leads to abortion or stillbirth, but congenital malformations have not been reported. Although spirochetes or their remnants may persist in the CNS or other sequestered sites after bacteremia has resolved, posttreatment sequelae and prolonged disability have not been not documented for relapsing fever or *B. miyamotoi* disease. Partial immunity against reinfection seems to develop in residents of areas with perennial elevated risk.

PREVENTION

There is no vaccine for LBRF, TBRF, or *B. miyamotoi* disease. Reduction of exposure to lice and ticks is the key strategy for prevention. LBRF can be prevented through improved personal hygiene, reduction of crowding, better access to hot water (≥60° C) for clothes washing, and selected use of pesticides. Clothing is an important factor in maintaining the human body louse. The risk of TBRF can be reduced by construction of houses with concrete or sealed plank floors and without thatched roofs or mud walls. Houses and cabins in forested areas pose a risk in western North America when rodents nest in the roof, attic, or wall spaces or under the structure. Buildings infested with *Ornithodoros* ticks can be treated with pesticides and then rodent-proofed. If residing in a high-risk environment, individuals should not sleep on the floor, and beds should be moved away from the wall. Individuals with recreational or occupational exposure to caves, where mammals, including bats, may reside, merit advice about the risk of TBRF. Following exposure at a site of TBRF risk, treatment with doxycycline (either a single dose of 100 mg or 200 mg on day 1 followed by 100 mg/d for 4 days) was efficacious in preventing infection in a placebo-controlled trial. Recommendations for preventing *B. miyamotoi* infection follow those for reducing risk of Lyme disease from exposure to the vector hard ticks (Chap. 186).

FURTHER READING

Barbour AG, Schwan TG: Borrelia, in *Bergey's Manual of Systematics of Archaea and Bacteria*. WB Whitman et al (eds). John Wiley & Sons, Inc., 2018, pp 1-22.

Binenbaum Y et al: Single dose of doxycycline for the prevention of tick-borne relapsing fever. Clin Infect Dis 71:1768, 2020.

Butler T: The Jarisch-Herxheimer reaction after antibiotic treatment of spirochetal infections: A review of recent cases and our understanding of pathogenesis. Am J Trop Med Hyg 96:46, 2017.

Christensen J et al: Tickborne relapsing fever, Bitterroot Valley, Montana, USA. Emerg Infect Dis 21:217, 2015.

Gugliotta JL et al: Meningoencephalitis from Borrelia miyamotoi in an immunocompromised patient. N Engl J Med 368:240, 2013.

Isenring E et al: Infectious disease profiles of Syrian and Eritrean migrants presenting in Europe: A systematic review. Travel Med Infect Dis 25:65, 2018.

Krause PJ et al: *Borrelia miyamotoi* infection in nature and in humans. Clin Microbiol Infect 21:631, 2015.

Moran-Gilad J et al: Postexposure prophylaxis of tick-borne relapsing fever: Lessons learned from recent outbreaks in Israel. Vector Borne Zoonotic Dis 13:791, 2013.

Nordmann T et al: Outbreak of louse-borne relapsing fever among urban dwellers in Arsi Zone, Central Ethiopia, from July to November 2016. Am J Trop Med Hyg 98:1599, 2018.

Salih SY, Mustafa D: Louse-borne relapsing fever: II. Combined penicillin and tetracycline therapy in 160 Sudanese patients. Trans R Soc Trop Med Hyg 71:49, 1977.

Schwan TG et al: Tick-borne relapsing fever and *Borrelia hermsii*, Los Angeles County, California, USA. Emerg Infect Dis 15:1026, 2009.

Telford SR et al: Blood smears have poor sensitivity for confirming *Borrelia miyamotoi* disease. J Clin Microbiol 57:e01468, 2019.

von Both U, Alberer M: Images in clinical medicine. *Borrelia recurrentis* infection. N Engl J Med 375:e5, 2016.

Warrell DA: Louse-borne relapsing fever (*Borrelia recurrentis* infection). Epidemiol Infect 147:e106, 2019.

Wormser GP et al: Borrelia miyamotoi: An emerging tick-borne pathogen. Am J Med 132:136, 2019.

186 Lyme Borreliosis

Allen C. Steere

DEFINITION

Lyme borreliosis is caused by a spirochete, *Borrelia* (also called *Borreliella*) *burgdorferi sensu lato*, that is transmitted by ticks of the *Ixodes ricinus* complex. The infection usually begins with a characteristic expanding skin lesion, erythema migrans (EM; stage 1, localized infection). After several days or weeks, the spirochete may spread to many different sites (stage 2, disseminated infection). Possible manifestations of disseminated infection include secondary annular skin lesions, meningitis, cranial neuritis, radiculoneuritis, peripheral neuritis, carditis, atrioventricular nodal block, or migratory musculoskeletal pain.